

IoT in Sports – Changing the Game and Athletes

Dr. Babji Srinivasan

Department of Applied Mechanics & Biomedical Engineering
IIT Madras

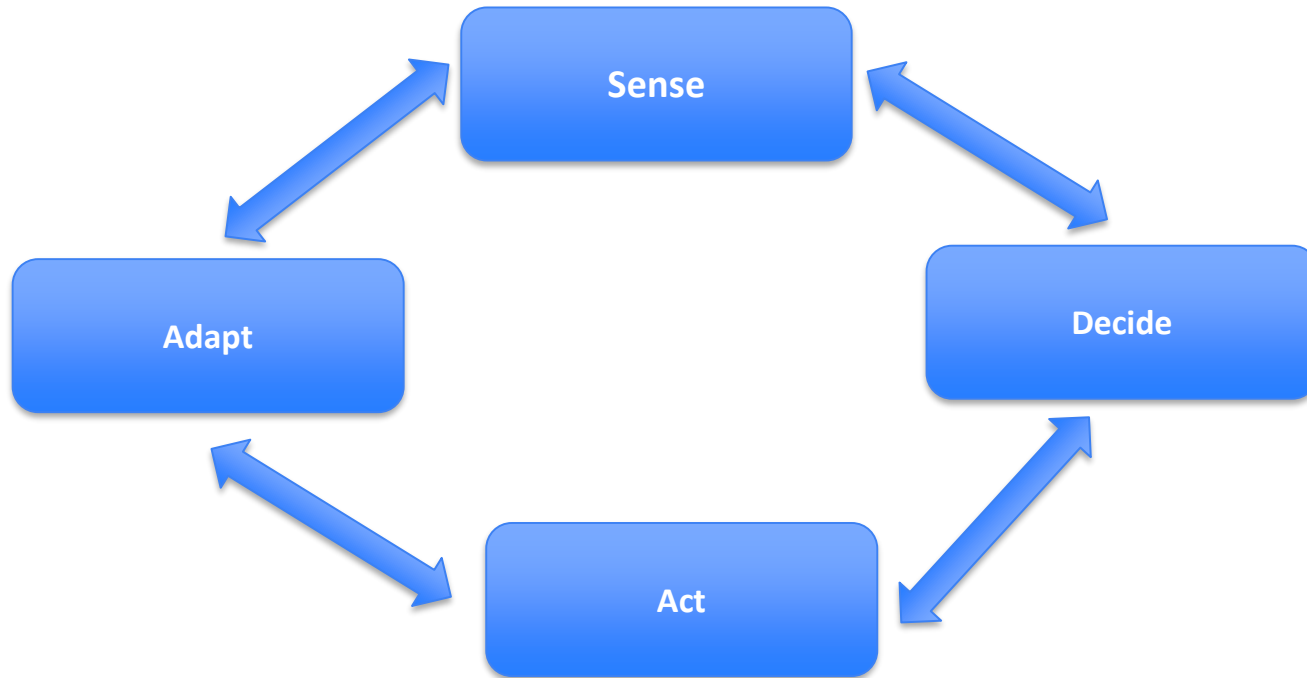
babji.srinivasan@iitm.ac.in



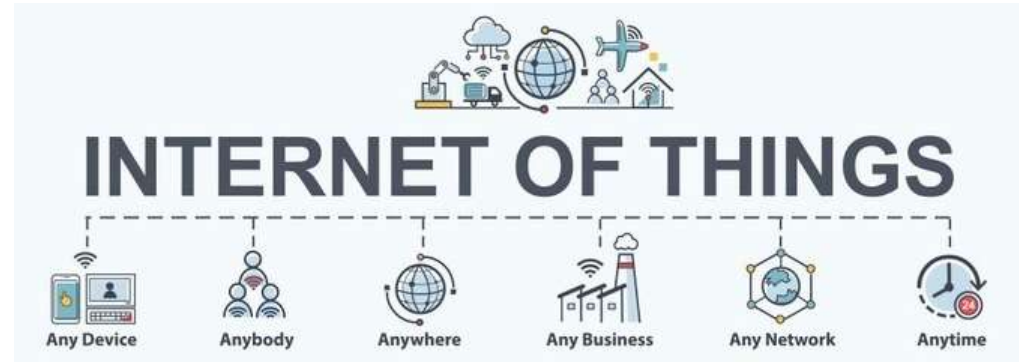
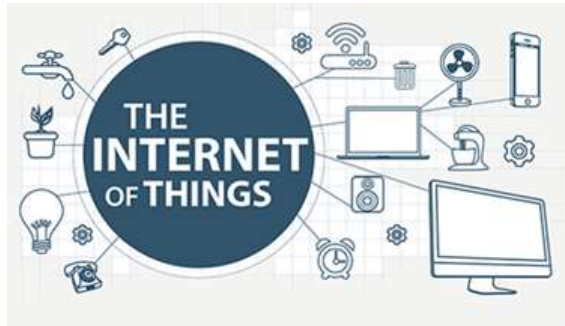
SMART SYSTEMS



WHAT IS A SMART SYSTEM



- Bridges the gap between physical and digital worlds



IoT is.... (According to EU)



- A global network infrastructure, linking physical and virtual objects through the exploitation of data capture and communication capabilities

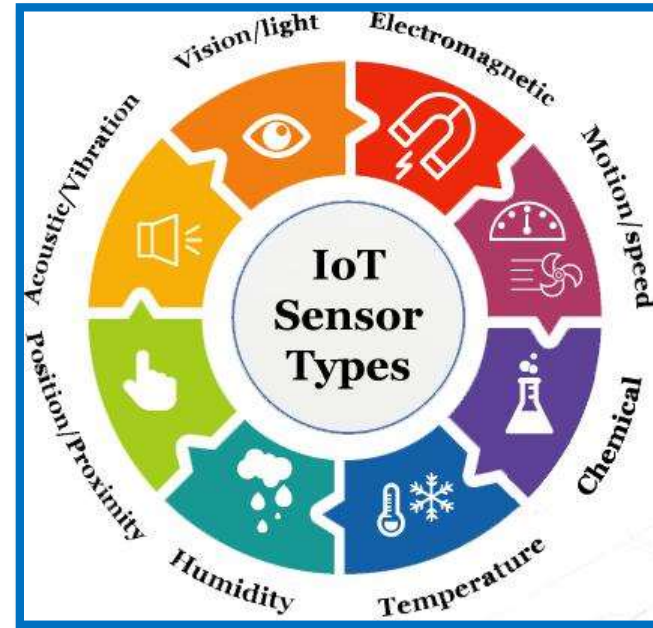


Building blocks of IoT

- End Devices / Nodes
- Gateways / Local Processing Nodes
- Connectivity
- Cloud Based Application and Storage

Building blocks of IoT

- End Devices / Nodes
 - Sensors
 - Objects
 - RFID Tags
 - Actuators



Building blocks of IoT

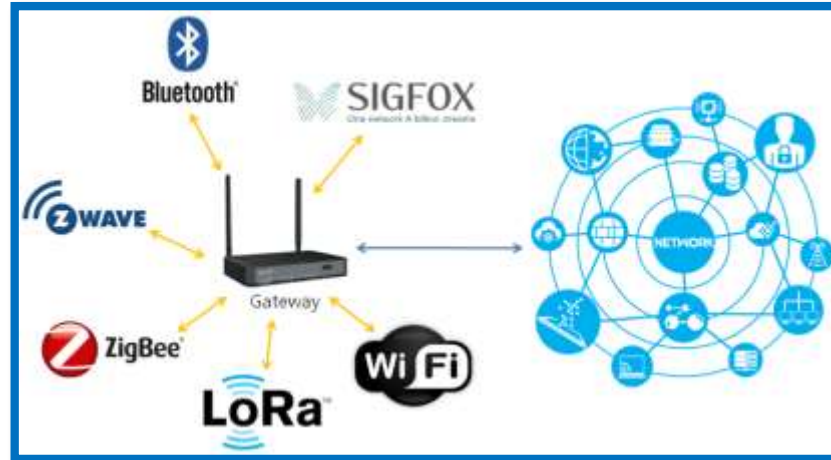
- Gateways / Local Processing Unit
 - Middleware
 - IoT Readers
 - Signal Receivers
 - Transceivers



Building blocks of IoT

- Network (Connectivity)

- WiFi
- Blue Tooth
- ZigBee
- LoRa



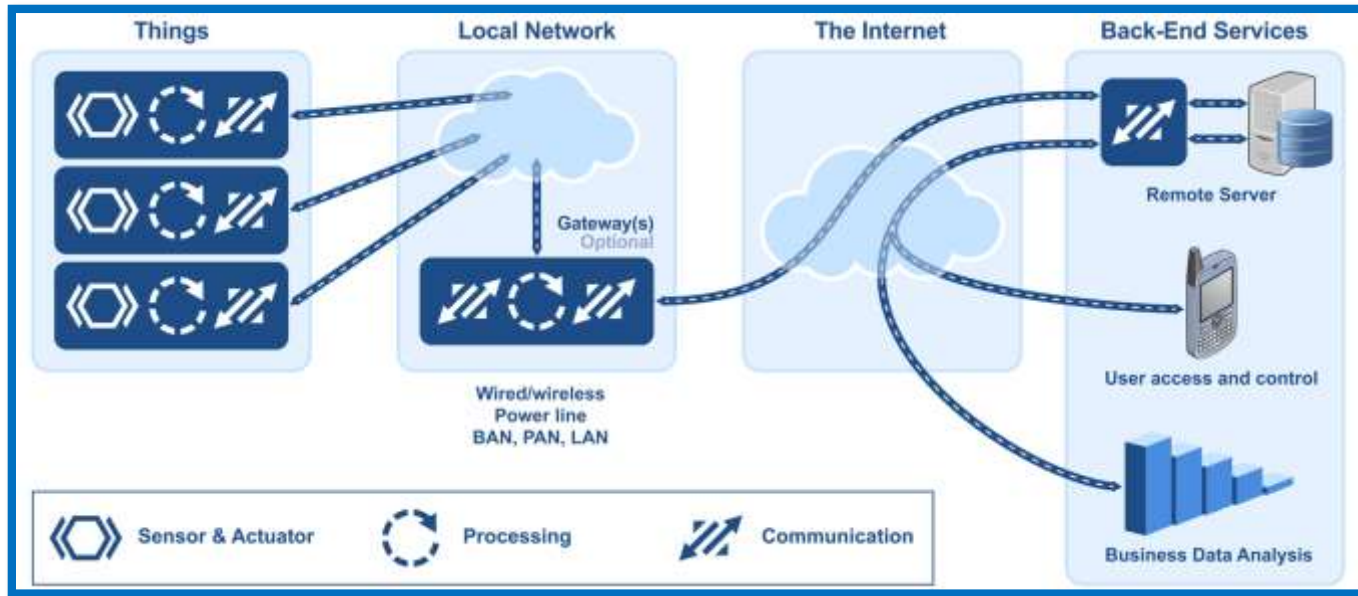
Building blocks of IoT

- Cloud based IoT Application and Storage

- Health
- Agriculture
- Traffic
- Smart City

Transport & Logistics  Fleet management, Goods tracking	Utilities  Smart metering, Smart grid management	Smart cities  Parking sensors, Waste management, etc.	Smart building  Smoke detector, Home automation
Consumers  Wearables Kids/senior tracker	Industrial  Process monitoring & control, Maintenance monitoring	Environment  Food monitoring/alerts, Environmental monitoring	Agriculture  Climate/agriculture monitoring, Livestock tracking

Building blocks of IoT



Layers of IoT



Applications Layer

- Industrial Applications: Aerospace, Heavy Machinery, Oil & Gas, Healthcare, Process Industries
- Consumer Applications: Smart home, Retail, Appliance, Wearables etc.



Management Layer

- Analytics, Data Management, BI, Visualization, Security, OSS/BSS, BRM/BPM
- Analysis of the data, decision making and business analytics as well as logical representation



Communication Layer

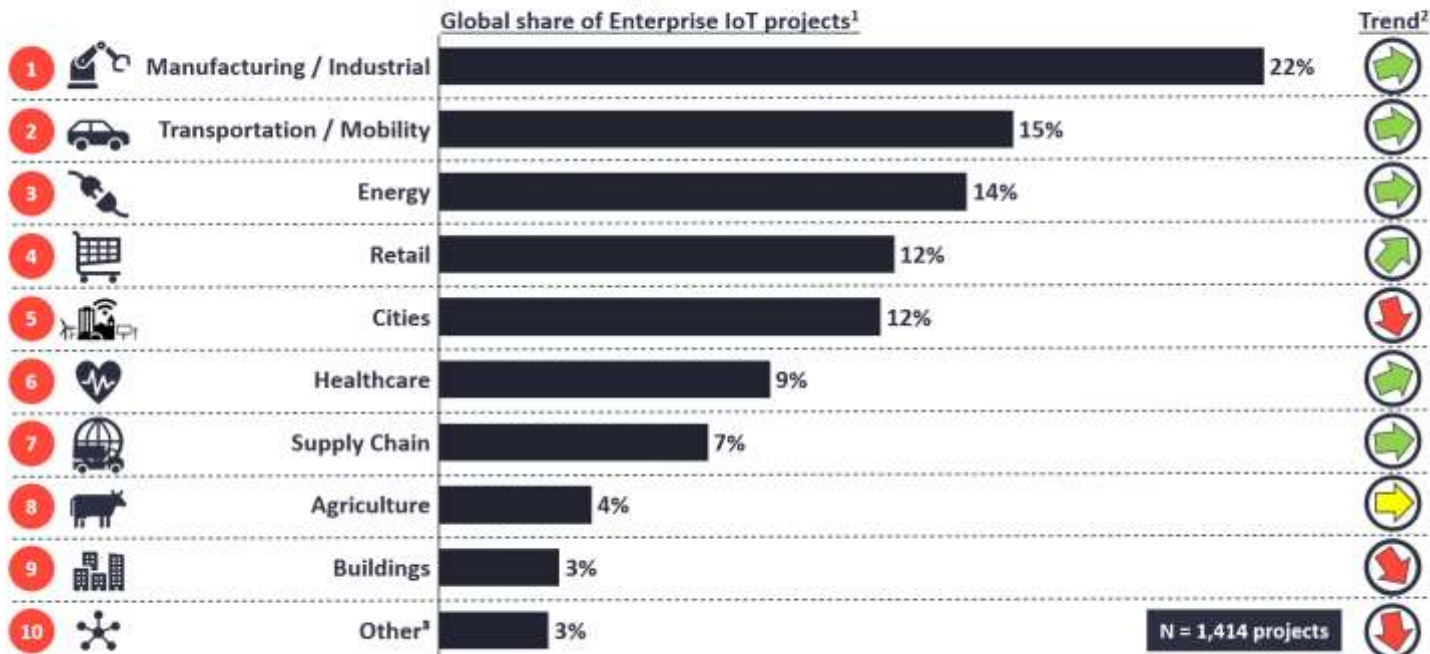
- Gateways: Encryption, SIM Module, Microcontroller, Signal Processors etc.
- Connectivity: Sensor network (LAN / PAN) Gateway Network (LAN / WAN)



Sensing Layer

- Measure, collect and process physical quantities in real time
- Gyroscope, GPS, IR, RFID, Solid State, Accelerometer etc.

Top 10 IoT Application areas 2020



Note: 1. Based on 1,414 publicly known IoT projects (not including consumer IoT projects eg smart home, wearables, etc.) 2. Trend based on relative comparison with % of projects in the 2018 IoT Analytics IoT project list e.g., a downward arrow means the relative share of all projects has declined, not the overall number of projects. 3. Other includes IoT projects from Enterprise & Finance sectors. Source: IoT Analytics Research - July 2020

Role of IoT in Sports

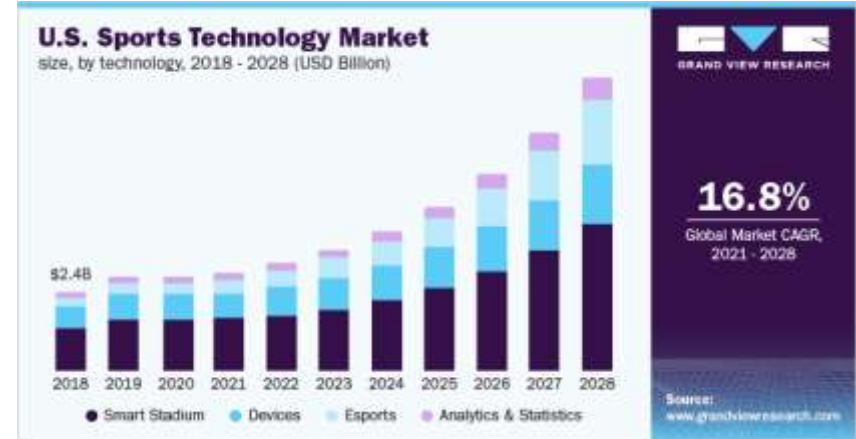
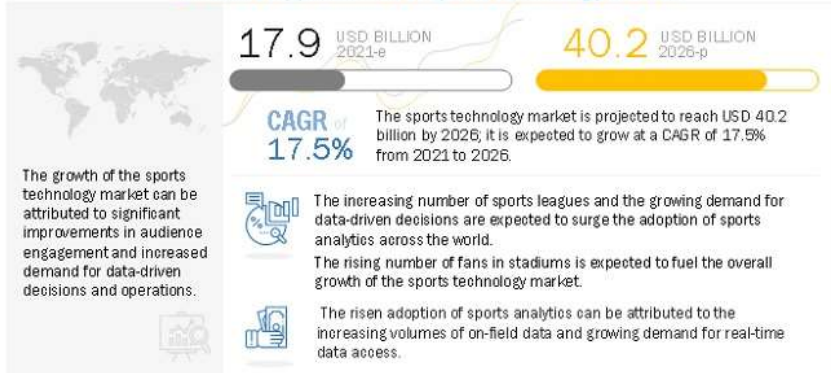
Athlete Health and Safety

Athlete/Team Performance

Smart Infrastructure



Attractive Opportunities in Sports Technology market



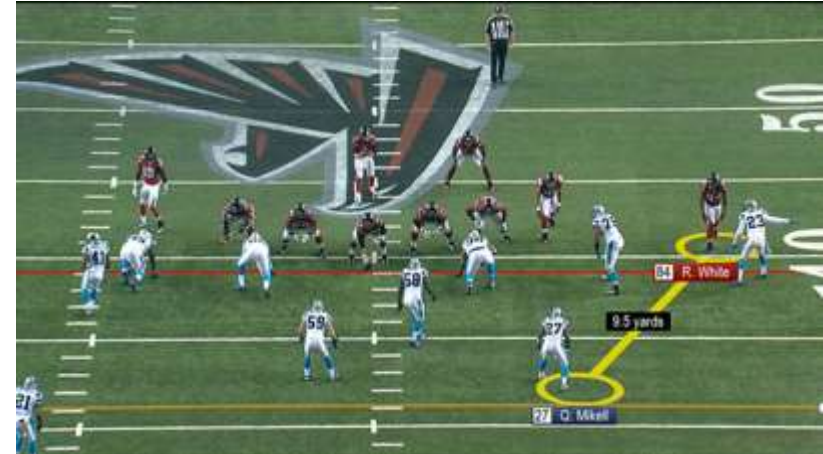
References:

[1] <https://www.globenewswire.com/en/news-release/2021/06/04/2241923/28124/en/The-17-9-Billion-Worldwide-Sports-Technology-Industry-is-Expected-to-Grow-at-a-CAGR-of-17-5-From-2021-to-2026.html>

- American Football: most popular sport in USA
- Football sensors:
 - Tracking spin rate, spiral efficiency and distance of the ball being thrown from the quarterback
 - Wilson's X Connected Football



- Players wear pads which contains many sensors
- Tracking of speed, acceleration, distance run and distance from other players in real time
- Useful for coaches to monitor athlete/team performance
- Used in National Football League (NFL)



- IoT-enabled FIFA authorized device
 - Catapult Player Smart Football tracker
 - records players' speed, total and sprint distance, and other metrics
 - Improves individual and team play



- Monitoring health of players
 - Use of smart helmets
 - Helmet sensors can easily alert the medical staff of any potential concussion
 - Convenient for medical staff to make difficult decisions of pulling a player from more potential harm



- IoT-enabled basketball
 - Tracking data regarding shots taken and shots made
 - Guided practice with better efficiency, accuracy, and trackability



- Limitations of standard fitness tracker:
 - Need to break strokes and check wrist
 - Interrupts the training session
- Solution:
 - Device attached to goggles
 - Instant audio feedback on performance





- Fitness trackers and smart-watches
 - Helps to remain productive
 - Analysis of actions and health over the entire day
 - Advanced measurements: blood oxygen levels, sleep tracking, ECG
- Better muscle training
 - Wearable pods to keep track of several important metrics



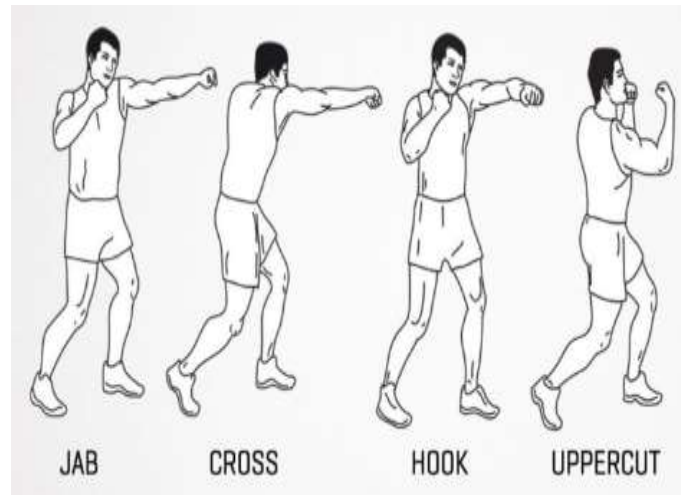


Sports science

Punch Type and range estimation

ML for boxing?

- In combat sports like boxing, **performance evaluation** is crucial for coaching and improvement.
- Machine learning (ML) techniques can accurately classify punch types (e.g., jab, hook, uppercut).
- Inertial Measurement Unit (IMU) sensors provide data for ML algorithms.
- Emphasize that punch classification alone is not enough to evaluate performance like tailoring training regimes, and enhancing offensive and defensive techniques.
- To address this aspect, we have focused on classifying punch types and ranges using various ML techniques trained on IMU sensor data.





Challenges

- Existing studies categorize punches based on execution, impact, and target distance.
- The **ideal punching position** for a boxer is when their weight is **evenly balanced** over both legs, enabling rotation around a central axis.
- To achieve this position, the boxer must move their **feet into the desired range** for punching.
- Being in the right range allows the boxer the opportunity to **deliver effective punches**.

Limited research on punch classification with corresponding ranges

Types of punch ranges



Long range



Long-Range Punches: Jabs, Straights, and Crosses
Control the fight's pace and maintain distance from the opponent

Mid Range



Mid-Range Punches: Hooks, Uppercuts, and Overhands
Create openings and closing the gap between the combatants

Close Range



Close-Range Punches: Body punches and Clinch punches
Power strikes delivered in close proximity to the opponent

Proposed methodology

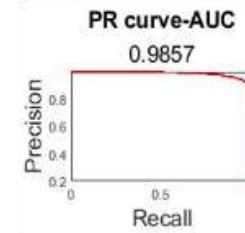
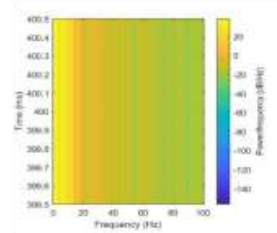
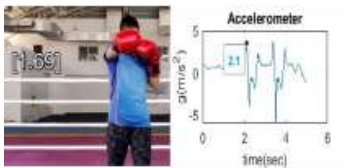
Data Acquisition (IMU data and video data)

Data Pre-processing (Synchronization)

Feature Extraction (Power spectrum density)

Machine Learning (classification)

- No punch
- Long-range rear jab to the head
- Mid-range rear hook to the head
- Mid-range rear uppercut to the head



Flowchart of proposed

What do you mean by feature extraction ? What is the purpose of feature engineering in machine learning?



Data Acquisition

- Elite boxers from Inspire Institute of Sports (IIS) in Karnataka, India and the use of IMU sensor data
- **Study Participants:** Eight elite boxers (6 males and 2 females) from IIS Extensive training and international experience
- Inertial Measurement Unit (IMU) Sensor: MetaMotion sensor with high-resolution and high-frequency data collection capabilities
- **Features:** 200 Hz sampling frequency, ± 16 g accelerometer, ± 2000 deg/s gyroscope
- **Real-Time Data Transmission:** Bluetooth capability for transmitting data to smartphones or computers
- Use of VelcroTM bands to secure the sensor underneath boxing gloves on both wrists



Data Acquisition

Data Collection: **Shadow boxing sessions** with 320 punches for each of the 14 different punch types

Punch Types: Long-range lead/rear jabs to head/body, long-range lead/rear hooks to head, mid-range lead/rear hooks to head, mid-range lead/rear steep hooks to head, mid-range lead/rear hooks to body, mid-range lead/rear uppercuts to head

Initial Classification: Four types of classification Categories: No punch, long-range rear jab to head, mid-range rear hook to head, mid-range rear uppercut to head

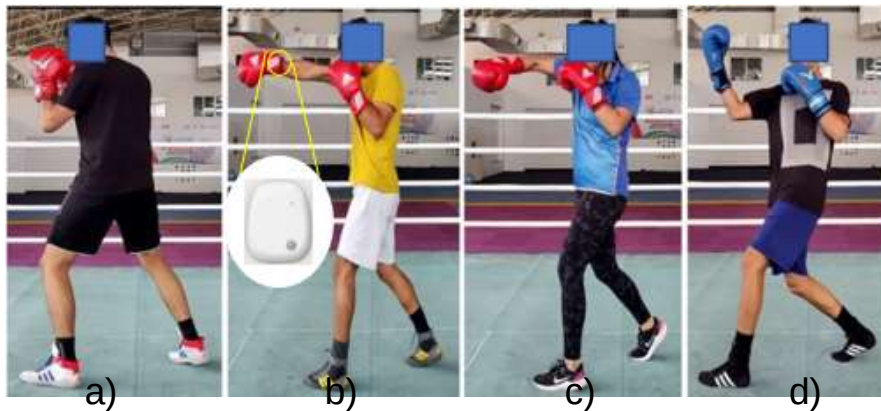


Figure a) No punch (b) long-range rear jab to head (c) mid-range rear hook to head (d) mid-range rear uppercut to head



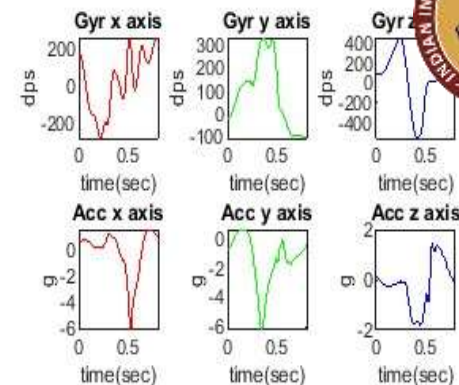
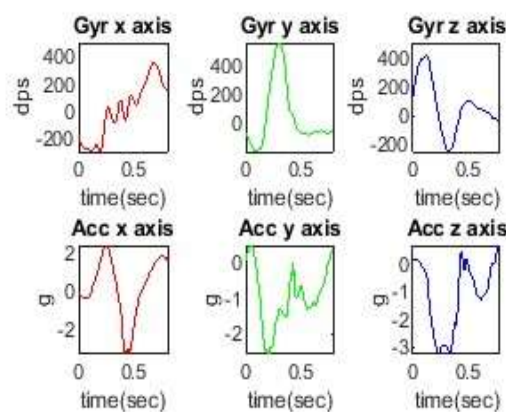
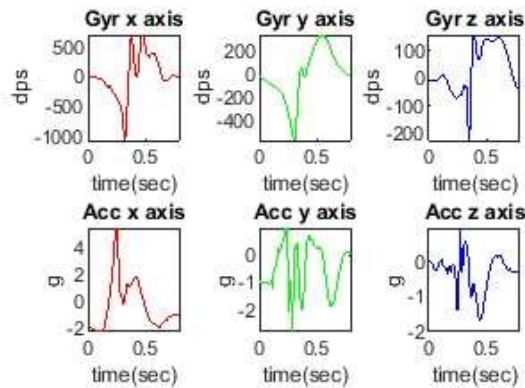
Data Pre-Processing and Feature extraction

- Time-Frequency Analysis and Feature Extraction for Punch Characterization
- Precise time synchronization and data processing techniques for accurate punch classification
- **Time Synchronization:** Ensuring accurate annotation and reliable ground truth information
- Integration of IMU data and video data for comprehensive analysis
- **Feature Extraction:** Utilizing IMU sensor data from accelerometer and gyroscope



Feature Engineering

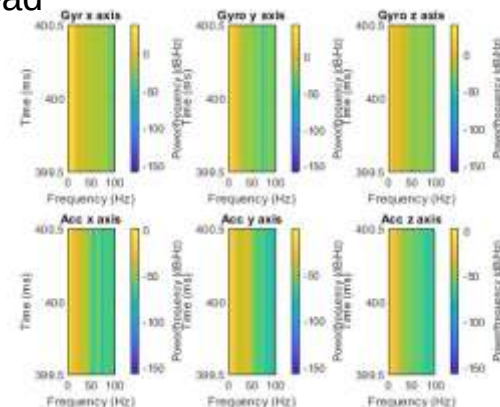
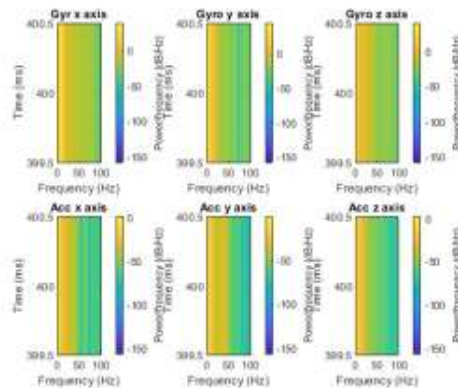
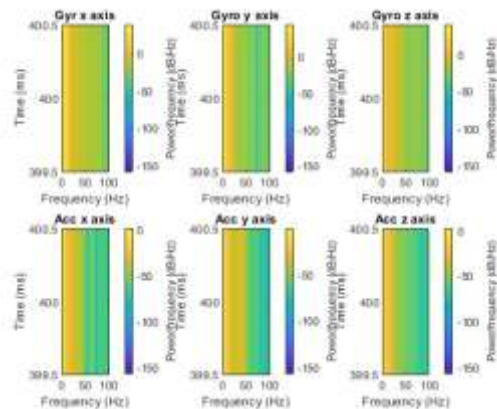
- Time-frequency analysis for the extracted spatio-temporal feature
- Power Spectral Density: Key feature extracted from the time-frequency analysis
- Characterizing Punches: Identifying patterns and trends through feature extraction.
- Helps associating the specific frequency ranges with punch types (jabs, hooks, uppercuts) and their respective ranges (long, mid, close)



6DOF long-range rear jab to head

6DOF mid-range rear hook to head

6DOF mid-range rear uppercut to head



Spectrogram of long-range rear jab to head

Spectrogram of mid-range rear hook to head

Spectrogram of mid-range rear uppercut to head



Supervised Machine learning models

- Dataset Split: Division of dataset into training and testing sets
- Training Set: 80% of the data, including N=250 punches from each category
- Testing Set: 20% of the data, comprising 50 punches from each category
- Ground Truth Labels: Obtained by visually inspecting the recorded video

Explain the difference between unsupervised and semi-supervised learning.

Machine Learning Models: Fine and Coarse Decision Trees, Random Forest, LDA, and QDA

- Selection based on literature
- Comparison of ML Technique Performance: Accuracy evaluation for punch type and range classification

What is the trade-off between bias and variance in machine learning models?



Performance Assessment

Model	Accuracy	Precision	Recall	F1-score
Fine Decision Tree	0.93	0.93	0.93	0.93
Coarse Decision Tree	0.60	0.60	0.74	0.66
Linear Discriminant	0.91	0.91	0.92	0.91
Quadratic Discriminant	0.65	0.65	0.78	0.71
Random forest	0.96	0.96	0.96	0.96



Discussion

- Random Forest: Highest accuracy among the tested algorithms (96 %)
- **Confusion Matrix** for Random Forest: Detailed assessment of classification accuracy (Figure)
- **Accurate classification** for most punch types, with some misclassifications observed for "no punch" and "rear hook" categories
- Statistical analysis of misclassified no punch category compared to true categories.
- Mean, standard deviation, and variance analyzed.

Random forest Confusion Matrix

True Class	No punch	45	1	3	1
	long range Jab		50		
	mid range Hook	1		48	1
	mid range Uppercut				50
		No punch	long range Jab	mid range Hook	mid range Uppercut
		Predicted Class			

Limitation

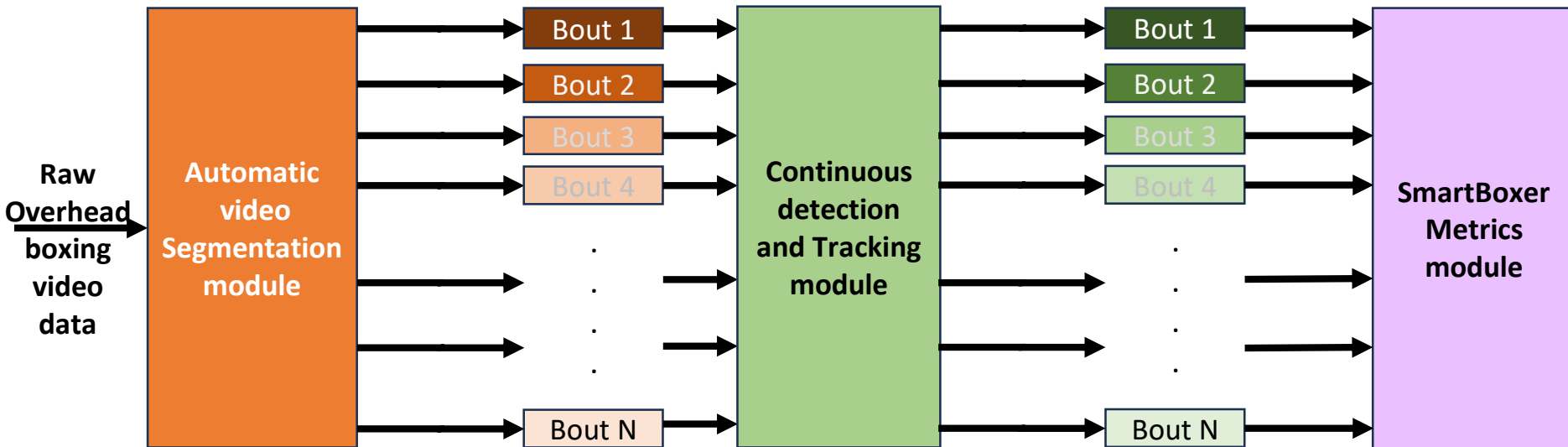
- Features of misclassified no punch category matched mid-range hook data instead of no punch.
- Free preparatory motion for a punch misclassified as a mid-range hook.



Sports science

SmartBoxer System

Overall workflow





Automatic video segmentation

Objective

To segment the raw unscripted training sessions into meaningful bouts

Challenges

- Overhead camera angle
- Multi-person detection
- Determine the start and the end
- Detecting outliers

Detection from top-view

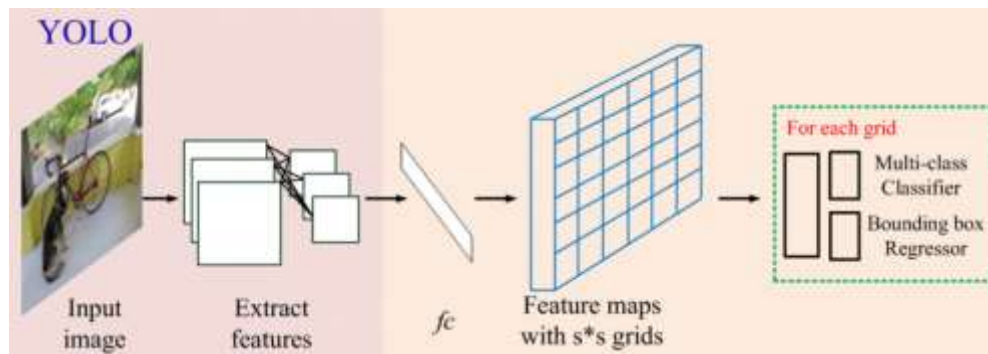
- Literature review suggested that the recent state-of-the-art object detection networks like YOLO, SSD performs better to that of R-CNN, Faster R-CNN, MASK R-CNN etc., in IMAGENET, COCO contests.
- Moreover, networks like YOLO, SSD are single-staged i.e., performs localization and classification at the same step compared to the two-staged networks.
- In our work, we chose YOLO to perform the base object (person) detection task from the overhead view video data.

So, What is YOLO ? How it works ? Why people are talking about it ?

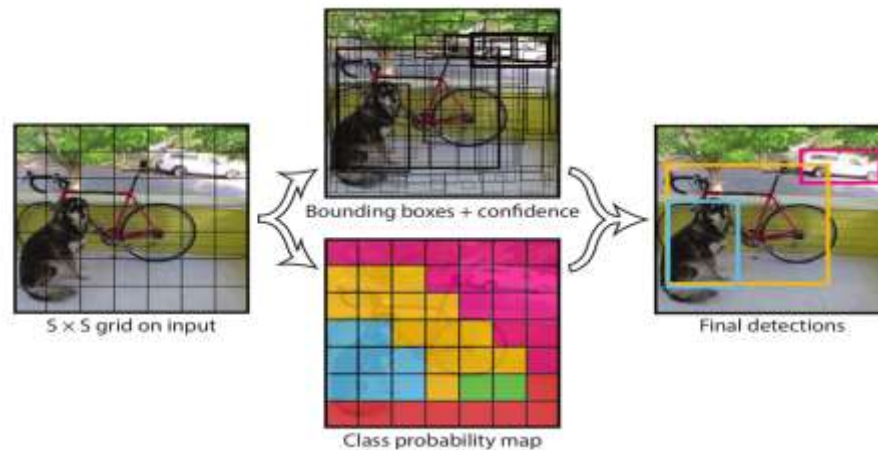
What is an activation function ? What is the main purpose of the activation function in a convolutional neural network (CNN)?

YOLO: You Only Look Once

Working of YOLO [Zhang hang 2019]



General illustration of YOLO
source:
<https://leimao.github.io/blog/YOLOs/>



It is one of the state-of-the-art detection model used for the detecting real-world objects in real-time. Developed by *Joseph Redmon*, to make the detection faster.

YOLO: You Only Look Once

➤ Features extraction module

- Darknet is 53 layered deep CNN model developed to act as a "backbone" feature extraction module for YOLO
- It uses convolutional kernels of sizes 1x1, 3x3.
- Residual (skip/shortcut) connections were also used in this network.
- The flattened FC layer will be used as feature representations of the image

Table 1 Performance against the state-of-the-art backbone CNN models

Backbone	Top-1	Top-5	Bn Ops	BFLOP/s	FPS
Darknet-19 [15]	74.1	91.8	7.29	1246	171
ResNet-101[5]	77.1	93.7	19.7	1039	53
ResNet-152 [5]	77.6	93.8	29.4	1090	37
Darknet-53	77.2	93.8	18.7	1457	78

What is the vanishing gradient problem in deep learning, and how can it be mitigated?

DarkNet -53 architecture

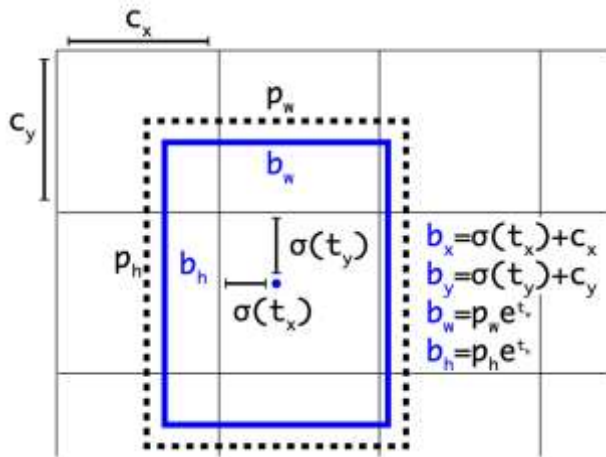
Type	Filters	Size	Output
Convolutional	32	3 × 3	256 × 256
Convolutional	64	3 × 3 / 2	128 × 128
1× Convolutional	32	1 × 1	
Convolutional	64	3 × 3	
Residual			128 × 128
Convolutional	128	3 × 3 / 2	64 × 64
Convolutional	64	1 × 1	
2× Convolutional	128	3 × 3	
Residual			64 × 64
Convolutional	256	3 × 3 / 2	32 × 32
Convolutional	128	1 × 1	
8× Convolutional	256	3 × 3	
Residual			32 × 32
Convolutional	512	3 × 3 / 2	16 × 16
Convolutional	256	1 × 1	
8× Convolutional	512	3 × 3	
Residual			16 × 16
Convolutional	1024	3 × 3 / 2	8 × 8
Convolutional	512	1 × 1	
4× Convolutional	1024	3 × 3	
Residual			8 × 8
Avgpool		Global	
Connected		1000	
Softmax			



YOLO: You Only Look Once

➤ Prediction module

- It looks at the image in **grids**. If the centre of the object falls within the grid, then that grid is responsible for that object. Each grid predicts (regress) bounding box $[t_x, t_y, t_w, t_h]$ and a confidence score.
- If the object is not entirely on any grid, then YOLO uses the offset (C_x, C_y) , and the prior bbox information



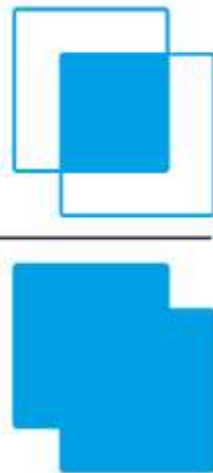
YOLO predicts an objectness score for each bounding box using logistic regression with sum of squared error loss !

YOLO: You Only Look Once

- This confidence score reflects how the **grid is confident** that the object.

$$CS = \Pr\{obj\} * IOU \quad (1)$$

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$



IOU is a threshold that decides how much the predicted area overlaps with the prior (GT).

- Each grid cell also predicts **conditional class probabilities**

$$COS = \Pr\left\{\frac{class}{obj}\right\} \quad (2)$$

- This probabilities are **conditioned on the grid cell that it contains the object**. Each grid predicts only one conditional class probabilities.

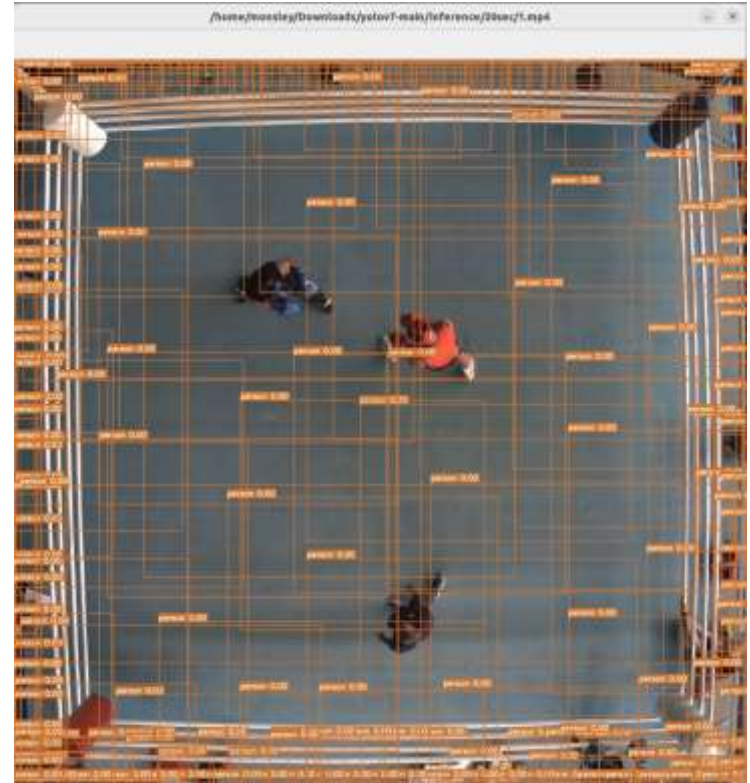
- Thus, each grid predicts the class that 'may' be there in the anchor box region by using independent logistic classifiers with binary cross-entropy loss function.
- During testing, this COS is multiplied with the confidence prediction.

Now that we have understood the base working of YOLO model. The recent models are just upgradation to this model. The recent YOLOv7 which we had used in this **SmartBoxer System** has the following upgradations, We have mentioned a few here.

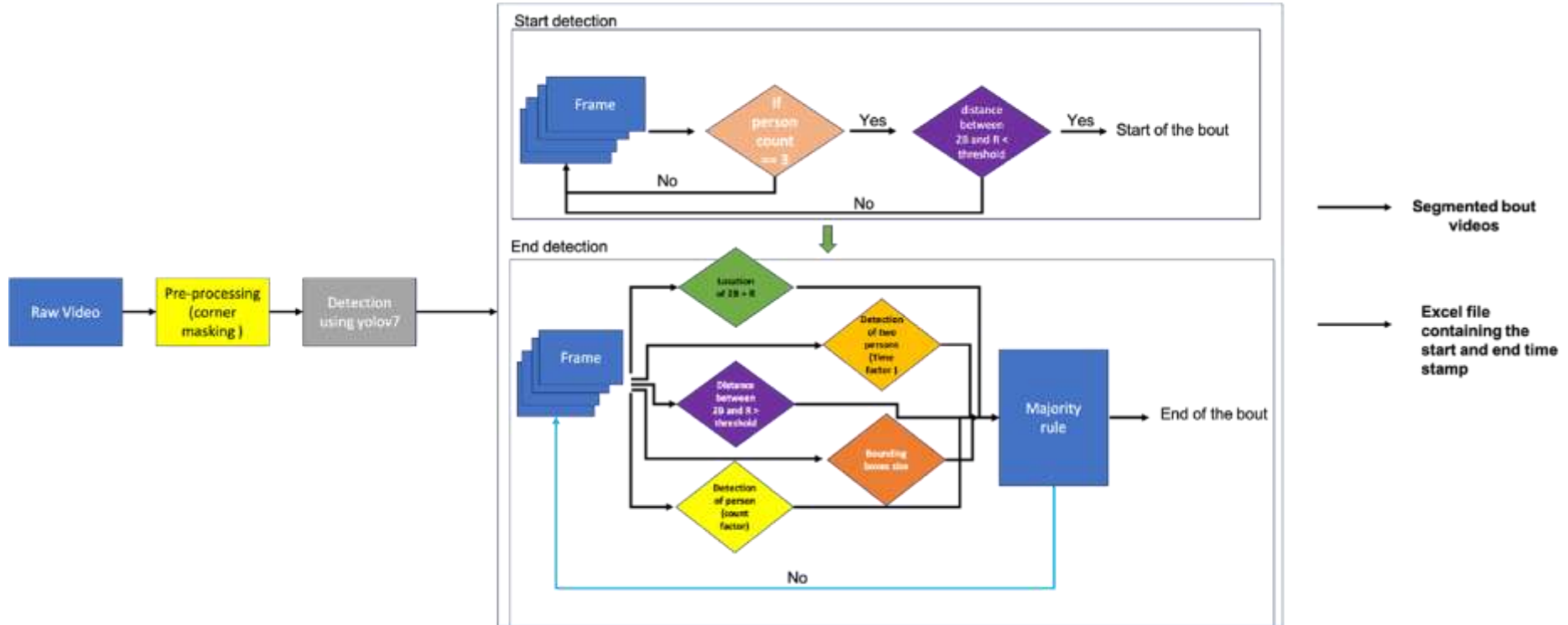
1. **Extended Efficient Layer Aggregation** - The shorter the gradient needs to travel, the more powerful the network will be able to learn. They used expand, shuffle, merge cardinality to achieve the ability to continuously enhance the learning ability of the network without destroying the original gradient path.
2. **Compound Scaling method** – Any increase in the depth of a computational layer should be balanced by increase in width of transitional layers

An illustration: YOLO inferenced on a frame

- To show how YOLO uses the grids to estimates the anchor box on a top-view boxing data.
- The confidence threshold was kept low (0) to highlight full extent of YOLO estimating the anchor boxes.
- Class was kept as 'person'



- Proposed AVS algorithm:



Pre-processing stage

- All external regions of the ring side are masked using black pixels.
- This step is performed to handle prevent incorrect detection of people standing near the ring.
- To prevent false detection of the poles by the pre-trained weight, we masked the poles also with black pixels.
- This processed frame is fed as input to YOLO network.



Module

Rules to determine the start of a bout

1. Detecting three person inside the ring for few seconds.
2. Distance between both the boxers and referee.

Rules to determine the end of a bout

1. Distance between both the boxers and referee.
2. Location of both the boxers and referee
3. Detecting two persons inside the ring for few seconds.
4. Considering size of the bounding boxes to overcome false detection of persons.
5. Detecting 0, 1, 2, 4, and 5 persons inside the ring for few seconds.

*These rules were established by observing more than 30 hours of collected data.
As a result, we would get the segmented bout videos



Continuous detection and tracking

- Objective

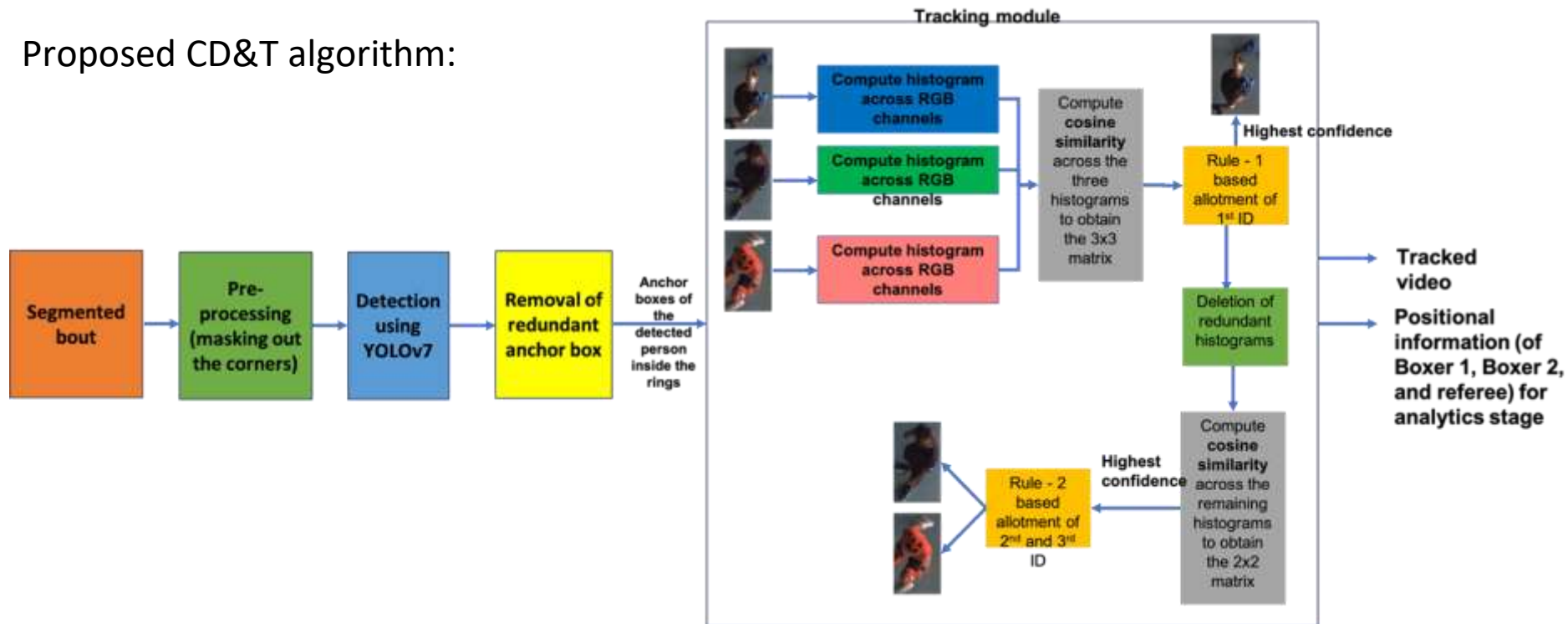
To detect and track the boxers inside the ring continuously

Challenges

- Unconventional camera angle
- Monocular vision
- Varying lighting conditions
- Presence of multiple athletes
- Occlusion
- Faulty detections

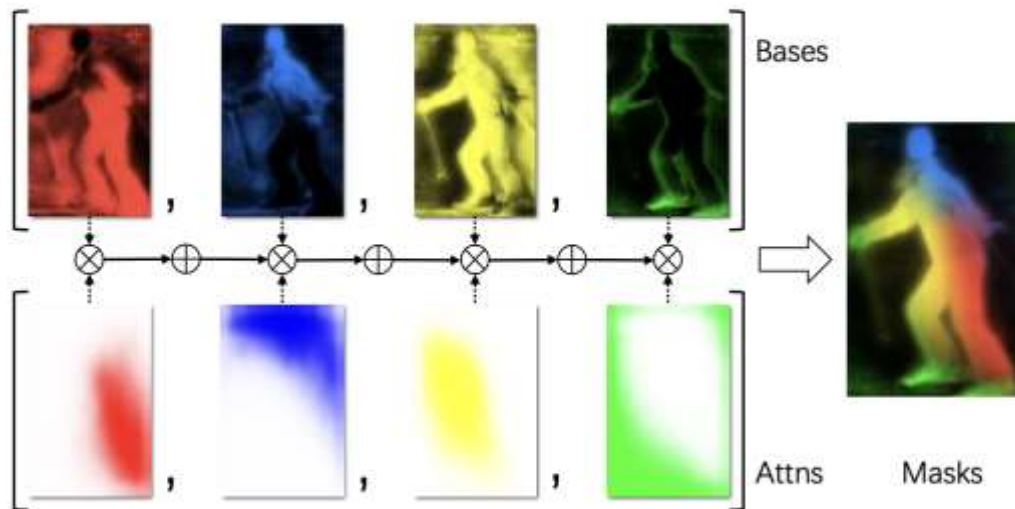
CD&T

Proposed CD&T algorithm:

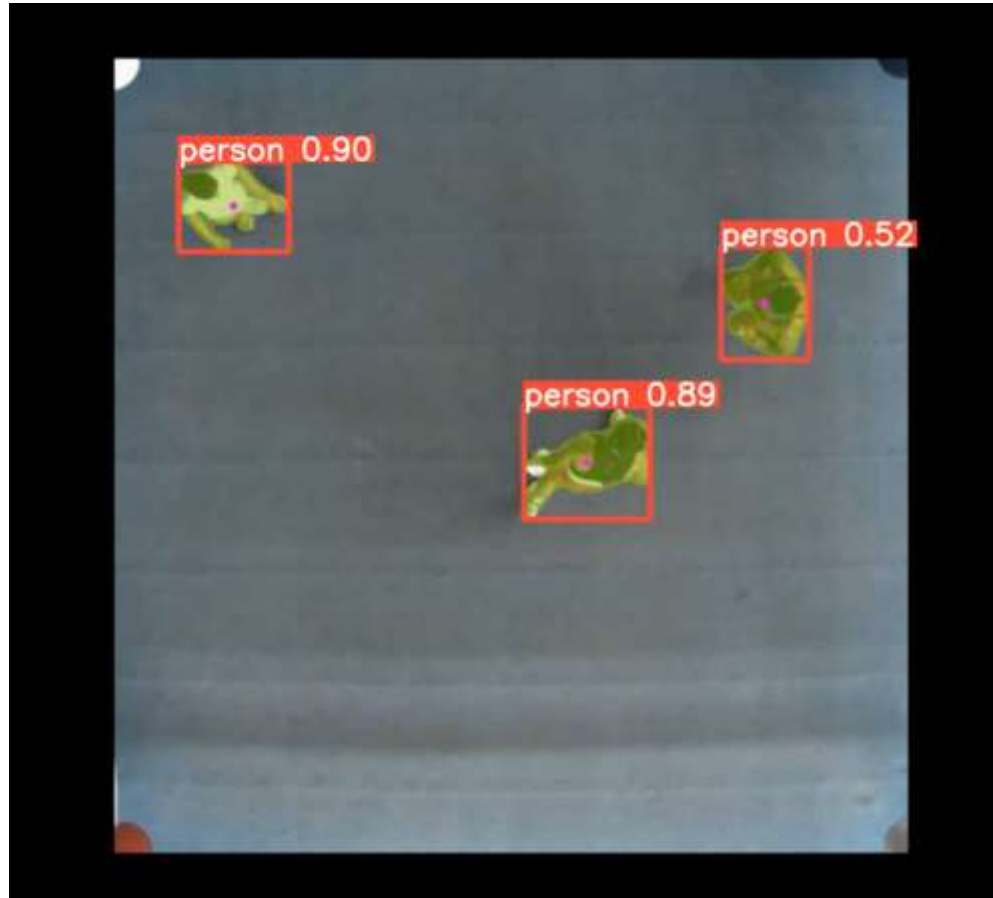


YOLOv7 {integrated with BlendMask (for segmentation)}

- From each of the detected bounding box, we segment the individuals using BlendMask.
- It's uses a fully convolutional network which predicts the confidence score maps.
- Also, it uses attention module to store the coarse and pose level information.
- The outputs from FCN and Attn modules are combined to get the segmentation masks.



Segmented output



When initial conditions are not met, move to the next frame

When initial conditions are met

Each channel in the segmented mask would be calculated.

The highest pixel range across the RGB channel is used to normalization. Helps in intra- and inter color discrimination

As we need to compare two frames, we temporarily store the histograms, till the above mentioned process gets completed for the succeeding frame

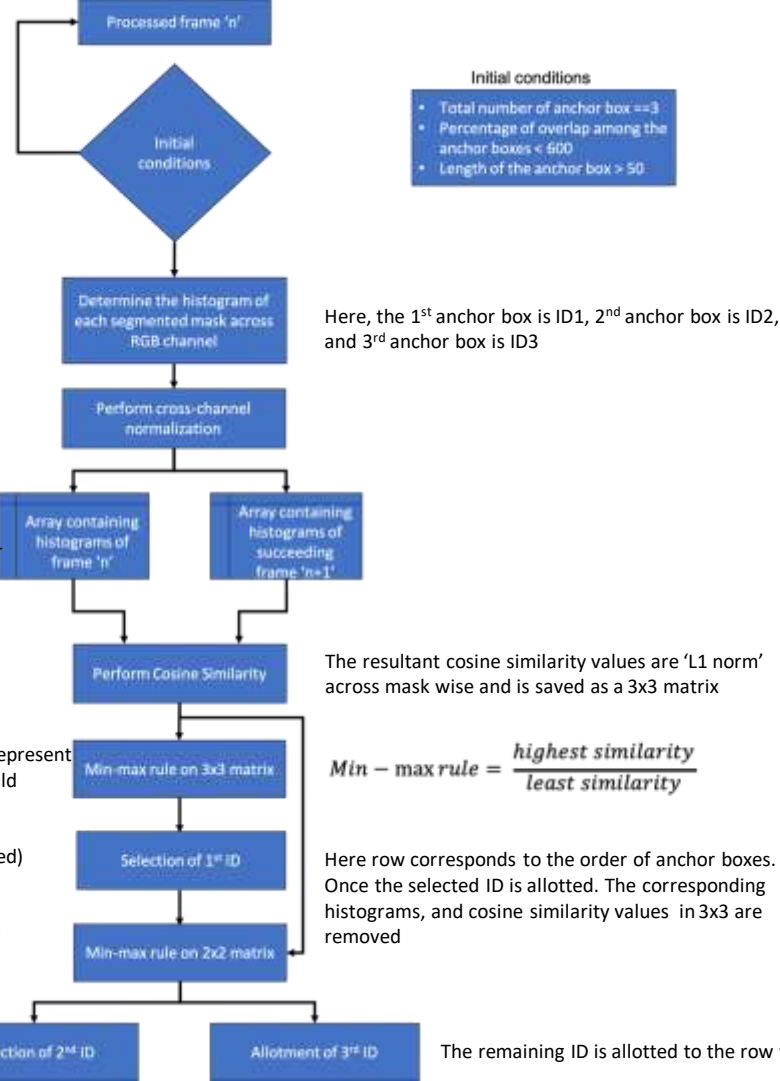
In total, we would compare 9 histograms of frame n against 9 histograms of frame n+1

This rule is applied along the row-wise (which represent segmentation wise) matrix. As a result, we would obtain a 3x1 matrix.

The first ID allocation is done to the row (selected) with the highest ratio.

The rule is applied on the remaining 2x2 matrix.

The second ID allocation is done to the row (selected) with the highest ratio.

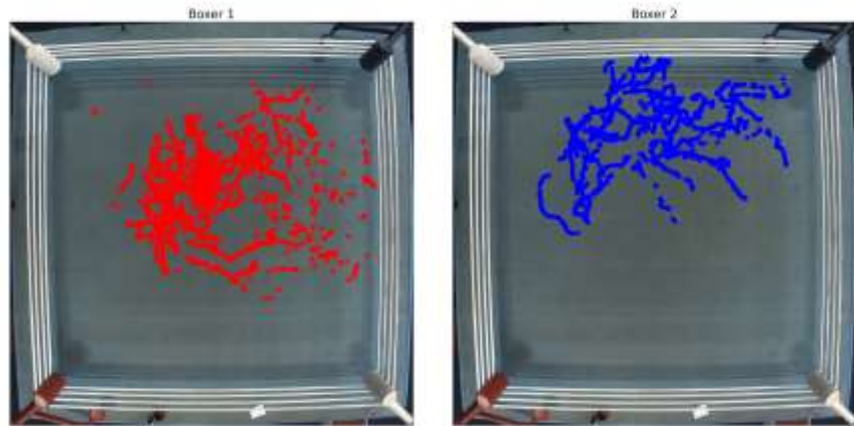


SmartBoxer Metrics module



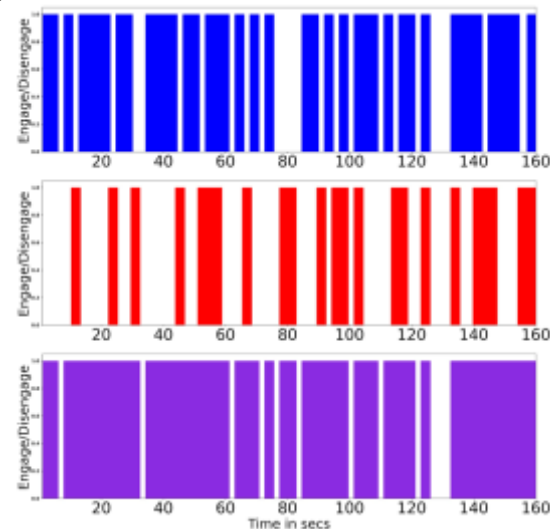
Hotspot:

This metric helps to reflect the players movement pattern in the ring



Engagement/disengagement:

This metric helps to highlight how often the boxer has engaged his opponent

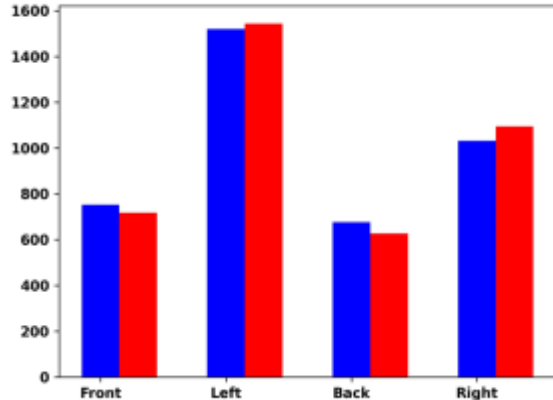


SmartBoxer Metrics module



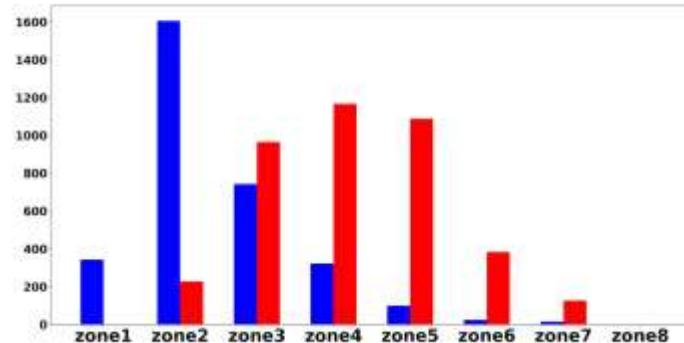
Directional histogram:

This metric helps to capture the players directional changes in the ring.



Zone management :

The ring was segregated into 8 strategic zones. The frequency/instances of boxers inside such zones gets reflected in this metric



Thank you for listening